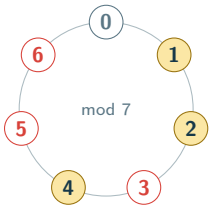
NUMBER THEORY ON ONE PAGE

The Legendre symbol and quadratic reciprocity

A square-root test modulo a prime, the law that swaps its top and bottom, and the run of primes behind -163.

1 / THE RULE AND THE PICTURE

For an odd prime p, the symbol (D/p) records whether D is a square modulo p



The symbol asks whether $x^2 \equiv D \pmod p$ has a solution x, where the unknown is squared and D is the required remainder:

$$\left(\frac{D}{p}\right) = \begin{cases} +1, & x^2 \equiv D \text{ has a nonzero solution (residue)}, \\ -1, & x^2 \equiv D \text{ has no solution (nonresidue)}, \\ 0, & p \mid D. \end{cases}$$

x	0	1	2	3	4	5	6
$x^2 \bmod 7$	0	1	4	2	2	4	1

The clock shows $1, 2, 4$ as residues (gold) and $3, 5, 6$ as nonresidues (red), so $(2/7) = (4/7) = 1$, $(3/7) = -1$ and $(7/7) = 0$.

2 / COUNTING / EXACTLY HALF ARE SQUARES

Each nonzero square has two roots, x and $-x$, so $(p-1)/2$ residues exist

p	nonzero residues	count
5	1, 4	2
7	1, 2, 4	3
11	1, 3, 4, 5, 9	5
13	1, 3, 4, 9, 10, 12	6

Modulo 7 the inputs 3 and $4 = -3$ both give 2, the inputs 2 and 5 both give 4, and 1 and 6 both give 1, so the 6 nonzero inputs land on 3 distinct squares and the other 3 remainders are nonresidues.

3 / TWO RULES / MULTIPLYING THE TOP, AND TESTING -1

The top entry multiplies like signs, and -1 depends only on $p \bmod 4$

a	b	ab	labels
R	R	R	$(+1)(+1) = +1$
R	N	N	$(+1)(-1) = -1$
N	R	N	$(-1)(+1) = -1$
N	N	R	$(-1)(-1) = +1$

So $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$, and for example $(15/7) = (3/7)(5/7) = (-1)(-1) = +1$.

Since $-1 \equiv p - 1$, the sign is $(-1)^{(p-1)/2}$, which is $+1$ for $p = 4k+1$ and -1 for $p = 4k+3$:

$$\boxed{\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}}$$

Modulo 5 we have $-1 \equiv 4 = 2^2$, so $(-1/5) = +1$, whereas modulo 7 the value 6 is absent from the squares and $(-1/7) = -1$. Euler's criterion $(a/p) \equiv a^{(p-1)/2} \pmod p$ gives both rules.

4 / RECIPROCITY / SWAPPING TOP AND BOTTOM

For distinct odd primes the two answers agree unless both primes are $3 \bmod 4$

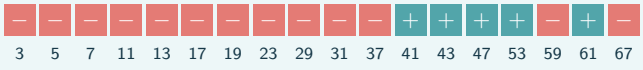
$$\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}} \left(\frac{p}{q}\right)$$

$p \bmod 4,\ q \bmod 4$	sign	relation	example
$1,1$ or $1,3$ or $3,1$	$+$	answers agree	$(5/13) = (13/5) = -1$
$3,3$	$-$	answers are opposite	$(3/7) = -1,\ (7/3) = +1$

The law turns a hard question into an easy one by reducing the bigger prime. For $D = -163$ the factor $163 \equiv 3 \pmod 4$ gives $(163/p) = (-1)^{(p-1)/2}(p/163)$, so $\left(\frac{-163}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{163}{p}\right) = (-1)^{p-1}\left(\frac{p}{163}\right) = \left(\frac{p}{163}\right)$: whether -163 is a square modulo p depends only on $p \bmod 163$, a square among 81 of the 162 classes, and primes spread evenly over those classes.

5 / AN UNEVEN START / WHY -163 BEGINS WITH ELEVEN NONRESIDUES

Every odd prime below 41 gives -1, and Euler's polynomial $n^2 + n + 41$ explains it



the sign of $(-163/p)$ for each prime p

Completing the square gives $4f(n) = (2n+1)^2 + 163$, so a prime dividing $f(n)$ makes -163 a square modulo that prime, which is impossible below 41. The forty values $f(0), \ldots, f(39)$ are all below 41^2 and so are prime, and $f(40) = 41^2$.

At $p = 41$ we have $-163 \equiv 1$, so the symbol turns $+1$. The deeper reason is that 163 is the largest Heegner number, so $\mathbb{Z}[\omega]$ with $\omega = \frac{1+\sqrt{-163}}{2}$ has unique factorisation and no element of prime norm below 41. With Mertens' $\sum_{p \le x} (\log p)/p = \log x + O(1)$ and the weighted imbalance $o(\log x)$, the nonresidue primes carry half the weight eventually.

Convention. p, q are distinct odd primes; R and N mean residue and nonresidue. After the student preamble *Squares, primes and 163*.